

Notes: Meghir and Pistaferri (Econometrica, 2004)

Income Variance Dynamics and Heterogeneity

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1 Big picture

- **Question.** Are individual earnings shocks really i.i.d., or do they display persistence, state dependence, and heterogeneity in risk across people and over time?
- **Why it matters.** Macroeconomic and household-finance models of savings, consumption, insurance, mobility, and poverty often rely on simple income processes. If income risk is time-varying rather than constant, those models can mismeasure precautionary behavior and welfare.
- **Core empirical challenge.** In panel earnings data, measured income dynamics mix together three objects: a permanent shock, a transitory shock, and measurement error. The paper's contribution is to separate them and then ask whether the variances of the permanent and transitory shocks themselves are predictable.
- **Main contribution.** The paper moves beyond constant-variance income processes and estimates education-specific ARCH-type models for the conditional variance of both permanent and transitory earnings shocks, allowing also for fixed unobserved heterogeneity in variances.
- **Data.** The analysis uses male heads from the PSID over calendar years 1967–1992, grouped into High School dropouts, High School graduates, and College graduates.
- **Main structural finding on the mean.** Earnings residuals are best described by a random-walk permanent component plus a transitory component with short persistence. The data reject the view that there are no permanent shocks.
- **Main variance finding.** The conditional variance of both the permanent shock and the transitory shock shows strong ARCH-type persistence. Risk is therefore state-dependent: a big shock today predicts higher volatility tomorrow.
- **Heterogeneity result.** The paper also finds evidence that some people permanently face more volatile earnings processes than others, even after conditioning on education, age, and year.
- **Most stable quantitative summary.** In pooled estimates, the persistence parameter in the variance is about 0.404 for the transitory shock and about 0.56 for the permanent shock. For High School graduates the point estimates are even larger.
- **Economic implications.** Relative to an i.i.d. income process, ARCH dynamics make poverty less persistent in the tails, because people hit by large shocks are more likely to experience another large move that takes them out of the tail. ARCH dynamics also raise the scope for precautionary savings by increasing higher-order income risk.
- **Bottom line.** Microeconomic earnings risk is not just large; it is also persistent in variance and heterogeneous across people. For quantitative work, treating income shocks as homoskedastic i.i.d. innovations is too simple.

2 Data and measurement (key objects)

2.1 Data source and sample construction

The data come from the 1968–1993 merged family and individual files of the PSID, which correspond to calendar years 1967–1992 because labor-income questions are retrospective.

The paper keeps:

- male household heads,
- ages 25 to 55,
- with at least 9 years of usable earnings observations.

The final estimation sample contains 2,069 individuals and 31,631 individual-year observations.

Interpretation. The long PSID panel is crucial. The later ARCH-identification arguments need many usable lags; short rotating panels would not be enough.

2.2 Earnings measure

The earnings variable is the head's labor income from all sources, including wages, bonuses, overtime, commissions, the labor component of farm and business income, and related labor receipts. Nominal earnings are deflated by the GNP personal consumption expenditure deflator, with 1992 as the base year.

Interpretation. The paper studies *annual real earnings*. It is therefore about medium-run household labor-income risk, not monthly wage volatility.

2.3 Education groups

Individuals are split into three education groups:

- **High School dropout:** fewer than 12 grades of schooling,
- **High School graduate:** at least a High School diploma but no College degree,
- **College graduate:** College degree or more.

The paper allows both the mean earnings process and the variance process to differ by education group.

Interpretation. Education is not just a control. It is a way to allow different labor-market segments to face different average growth, different shock composition, and different risk dynamics.

2.4 The conditional mean specification

For each education group, log real earnings are first regressed on year dummies, a quadratic in age, race, region, and an SMSA indicator. The paper allows these observable returns to vary flexibly with education, and the year effects capture education-specific aggregate shocks.

This first-stage residual is the object whose stochastic structure is studied.

Interpretation. The paper strips out predictable life-cycle and demographic variation first, then studies the unexplained part as the earnings-risk object of interest.

2.5 Core latent components of earnings risk

The unexplained component of earnings is decomposed into:

- **measurement error** r_{it} , assumed classical in the baseline discussion,
- **transitory shock** e_{it} , allowed to be serially correlated,
- **permanent component** p_{it} , which evolves as a martingale.

The observed growth residual g_{it} is therefore a mixture of all three.

Interpretation. This decomposition is the central measurement idea of the paper. The econometric problem is not just “how risky is earnings?” but “which type of risk is this, and how does its variance move over time?”

2.6 What Figures 1 and 2 show before any formal estimation

Figure 1 plots the mean of log real earnings over time. Figure 2 plots the variance of log real earnings over time, both for the pooled sample and by education group.

The raw patterns are:

- mean earnings are higher for more educated workers;
- cross-sectional earnings variance is highest for the least educated;
- earnings inequality rises noticeably during the sample, especially around the late 1970s and early 1980s.

Interpretation. The descriptive graphs already show that both mean levels and dispersion are not stable. The rest of the paper asks whether that changing dispersion reflects permanent shocks, transitory shocks, measurement error, or genuine variance dynamics.

2.7 Why measurement error matters

The unconditional variance of the transitory shock, the variance of measurement error, and the MA parameter of the transitory component are not separately identified from the autocovariances of earnings growth alone.

The paper therefore uses two complementary approaches:

- **bounds** implied by nonnegativity restrictions on variances,
- an **external estimate** of measurement error based on validation-study evidence suggesting that roughly 25% of earnings-growth variance is due to measurement error.

Interpretation. Measurement error is not a side issue. If ignored, it can make the transitory component look more important and can blur the distinction between transitory risk and noise.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline mean equation

The conditional mean of log earnings is written as

$$y_{it} = m_t^e + \beta_t^e Z_{it} + u_{it}, \quad (1)$$

where y_{it} is log real earnings, m_t^e is an education-specific year effect, Z_{it} is the vector of observable characteristics, and u_{it} is the unexplained component.

Interpretation. The paper first removes predictable changes in earnings due to observable demographics and aggregate year effects, then analyzes the dynamics of u_{it} .

3.2 Permanent, transitory, and measurement-error decomposition

The unexplained component is decomposed as

$$u_{it} = r_{it} + e_{it} + p_{it}, \quad (2)$$

with the permanent component evolving as

$$p_{it} = p_{i,t-1} + \zeta_{it}. \quad (3)$$

The transitory component follows an MA(q) process,

$$e_{it} = \Theta^e(L, q)\varepsilon_{it} = \varepsilon_{it} - \sum_{j=1}^q \theta_j \varepsilon_{i,t-j}. \quad (4)$$

Taking first differences of the residual gives

$$g_{it} \equiv \Delta u_{it} = \Delta r_{it} + \Theta^e(L, q)\Delta \varepsilon_{it} + \zeta_{it}. \quad (5)$$

Interpretation. Earnings growth residuals are informative because differencing removes fixed permanent levels, while the remaining autocovariances reveal whether the transitory component is MA(0), MA(1), and so on.

3.3 Determining the order of the transitory process

If the transitory component is MA(q), then the autocovariances of g_{it} above order $q + 1$ should be zero. The paper estimates these autocovariances and tests the restrictions.

The data imply that autocovariances are clearly nonzero up to order 2 but become small and statistically insignificant at order 3 and above. This leads the paper to impose

$$q = 1. \quad (6)$$

Interpretation. The key empirical message is that transitory earnings shocks are not white noise. They have short persistence, but the persistence does not last beyond one year in the MA representation.

3.4 Identifying the unconditional variance of permanent shocks

The variance of the permanent shock is identified from a symmetric sum of local autocovariances:

$$E \left[g_{it} \sum_{j=-(1+q)}^{1+q} g_{i,t+j} \right] = E(\zeta_{it}^2). \quad (7)$$

The paper estimates this by equally weighted minimum distance.

Interpretation. This is the paper's cleanest identifying result. By summing the right set of neighboring autocovariances, the transitory MA component cancels out and what remains is the permanent innovation variance.

3.5 What is and is not identified for the transitory component

For the transitory component, the variance of ε_{it} , the variance of measurement error r_{it} , and the MA coefficient θ are not jointly point-identified from the earnings-growth autocovariances alone.

The paper therefore reports:

- bounds based on the requirement that the variance of measurement error cannot be negative,
- point estimates that use an external value for the measurement-error variance.

Interpretation. The paper is explicit about partial identification here. It does not pretend that the data alone can fully separate transitory risk from measurement error.

3.6 ARCH model for the conditional variance

The paper models the conditional variances of the transitory and permanent shocks as ARCH(1) processes with time effects and individual heterogeneity:

$$E_{t-1}(\varepsilon_{it}^2) = \kappa_t^e + \gamma^e \varepsilon_{i,t-1}^2 + \lambda_i^e, \quad (8)$$

$$E_{t-1}(\zeta_{it}^2) = \kappa_t^\zeta + \varphi^\zeta \zeta_{i,t-1}^2 + \lambda_i^\zeta. \quad (9)$$

Interpretation. These are the central variance equations. A positive γ^e or φ^ζ means volatility clusters: a large shock today predicts a larger conditional variance tomorrow. The λ_i terms allow some individuals to have permanently more volatile earnings paths.

3.7 Moment conditions for the transitory and permanent variances

For the transitory variance, the paper derives an orthogonality condition of the form

$$E_{t-2}[g_{i,t+q+1}g_{it} - \theta_q \kappa_t^e - \gamma^e g_{i,t+q}g_{i,t-1} - \theta_q \lambda_i^e] = 0. \quad (10)$$

For the permanent variance, the key orthogonality condition takes the analogous form

$$E_{t-q-3} \left[g_{it} \sum_{j=-(1+q)}^{1+q} g_{i,t+j} - \varphi^\zeta g_{i,t-1} \sum_{j=-(1+q)}^{1+q} g_{i,t+j-1} - \lambda_i^\zeta \right] = 0. \quad (11)$$

Interpretation. The paper does not observe ε_{it} and ζ_{it} separately. Instead, it creates products and sums of observed residual growth that behave like proxies for the latent conditional variances.

3.8 Within-groups estimating equation

With $q = 1$, the final within-groups GMM estimating equation is written as

$$\phi_{it} = \tau_t + \beta_1 \text{age}_{it} + \beta_2 \text{age}_{it}^2 + \gamma \phi_{i,t-1} + \nu_{it}. \quad (12)$$

For the transitory variance case,

$$\phi_{it} = g_{i,t+2} g_{it}, \quad (13)$$

and for the permanent variance case,

$$\phi_{it} = g_{it}(g_{i,t-2} + g_{i,t-1} + g_{it} + g_{i,t+1} + g_{i,t+2}). \quad (14)$$

Interpretation. This makes the ARCH problem look like a dynamic panel-data problem in observable objects. The coefficient γ is either the transitory ARCH coefficient γ^e or the permanent ARCH coefficient φ^ζ .

3.9 Alternative specifications the paper checks

The paper also considers richer mean specifications, including models with time-varying factor loadings or additive heterogeneity in earnings growth. The broad conclusion is that these alternatives do not overturn the preferred specification.

A particularly important test asks whether a quasi-differenced representation implies a coefficient on lagged earnings equal to one. The data do not reject this restriction, which supports the martingale permanent-shock setup.

Interpretation. The paper tries hard to show that the ARCH results are not just an artifact of a misspecified mean process.

4 Identification and instruments (what makes the estimates credible)

4.1 Why permanent shocks are identified cleanly

The permanent-shock variance is identified without requiring covariance stationarity. That is an important advantage because covariance stationarity is often rejected in earnings data.

The identifying moment for $E(\zeta_{it}^2)$ is also robust to serial correlation of the MA type in the transitory component and robust to classical measurement error.

Interpretation. This is a major strength of the paper. The most policy-relevant object—permanent income risk—does not depend on fragile stationarity assumptions.

4.2 Why the ARCH coefficients are identified

A notable insight of the paper is that identifying the ARCH coefficients does *not* require knowing the exact MA parameters of the transitory component. It only requires knowing the order q of that process.

The paper also shows that the ARCH coefficients remain identifiable even in the presence of classical or serially correlated measurement error, so long as the measurement error itself does not have an ARCH component.

Interpretation. The message is that the difficulty is not the latent-state structure per se. Once the transitory process is known to be MA(1), the higher-order products of the residuals contain enough information to recover variance dynamics.

4.3 Why the paper prefers within groups to first differences

Removing individual heterogeneity by first differences leads to weak instruments because the transformed lagged products are hard to predict from deeper lags.

The within-groups transformation is preferable when the time dimension is long enough. In this sample, individuals contribute between 9 and 26 time-series observations, making within-groups bias less concerning than weak-instrument bias.

Interpretation. The paper explicitly chooses the estimator based on finite-sample econometrics, not just algebraic convenience.

4.4 Instrument choice

The instruments are lagged products of residual-growth terms, constructed in the Arellano–Bond spirit but truncated to the first two available lags to avoid overfitting the reduced forms and to preserve power in overidentification tests.

For the transitory variance case, the useful instruments are deeper lags such as products dated $t - 2$ and earlier. For the permanent variance case, even deeper lagged sums of autocovariances are used because the identifying orthogonality condition itself is longer-horizon.

Interpretation. Because the latent shocks are not directly observed, the instruments are not raw lagged income growth. They are lagged *products* of income growth designed to line up with the variance moments.

4.5 Table IV: reduced-form relevance of the instruments

Table IV is the key identification diagnostic. It shows that in the first-difference specification the excluded instruments are often weak, especially for the permanent-shock ARCH model and for some education groups. In contrast, the within-groups reduced forms are strongly significant across essentially all cases.

Interpretation. This table is why the paper ultimately relies on within-groups GMM. The preferred estimator is not just theoretically elegant; it is the one with actual reduced-form power.

4.6 Overidentification and robustness

The Sargan-type overidentifying-restrictions tests are mostly acceptable. The one problematic case is the permanent-shock variance for College graduates, where the p-value is low in the baseline specification. When the instruments are lagged one period further, the p-value improves and the main coefficient remains statistically indistinguishable from the baseline result.

Interpretation. The problematic cell is acknowledged rather than hidden, and the paper shows that the substantive conclusion does not hinge on it.

4.7 Attrition

The identifying assumption on attrition is that it depends only on fixed observed or unobserved characteristics, not on realized shocks themselves. The paper checks sensitivity by restricting to individuals who remain in the PSID for 16 years or more. The resulting ARCH estimates are somewhat smaller but statistically similar.

Interpretation. Attrition may bias the variance persistence upward somewhat, but not enough to overturn the qualitative results.

5 Tables and figures

5.1 Figures 1 and 2: raw earnings levels and raw earnings dispersion

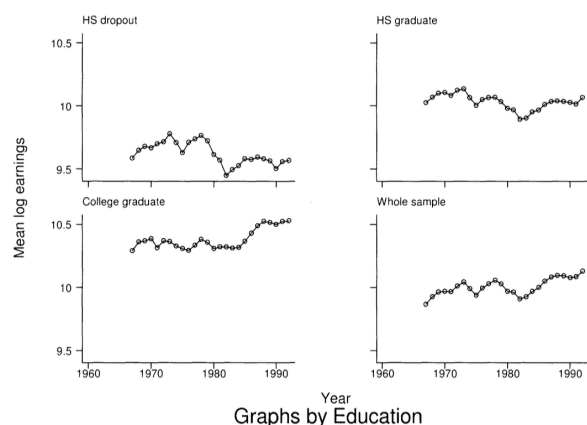


FIGURE 1.—The mean of log real earnings.

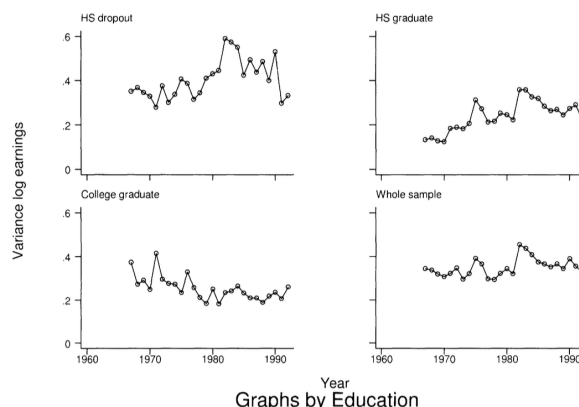


FIGURE 2.—The variance of log real earnings.

Read:Figure 1 shows average log real earnings rising strongly with education. Figure 2 shows that cross-sectional variance is highest for the least educated, with substantial time variation for all groups.

Economic message. Before any structural modeling, the data already say two things. First, there is strong heterogeneity in average earnings levels across education groups. Second, earnings dispersion itself moves over time, so homoskedasticity is not going to be a good approximation.

5.2 Tables I and II: the autocovariance pattern implies an MA(1) transitory component

TABLE I
THE AUTOCOVARIANCES OF THE UNEXPLAINED GROWTH OF EARNINGS^a

Order	Pooled Sample	High School Dropout	High School Graduate	College Graduate
0	0.109 (0.0051)	0.164 (0.014)	0.103 (0.0068)	0.065 (0.0058)
1	-0.0303 (0.0021)	-0.0535 (0.0055)	-0.028 (0.0027)	-0.0105 (0.0018)
2	-0.0079 (0.0014)	-0.0134 (0.0046)	-0.0077 (0.0015)	-0.0025 (0.0013)
3	-0.0024 (0.0011)	-0.0018 (0.0034)	-0.0026 (0.0014)	-0.0024 (0.0011)
4	0.0007 (0.0011)	0.0074 (0.0035)	-0.0015 (0.0013)	-0.0017 (0.0012)

^aAsymptotic standard errors are reported under the coefficient estimate. Values are pooled over all years and individuals.

TABLE II
TESTS OF THE HYPOTHESIS OF ZERO AUTOCOVARIANCES IN EARNINGS GROWTH^a

	Order of Autocovariance			
	1	2	3	4
Pooled Sample	320 df 24; $p=0$	61.2 df 23; $p=0$	27.7 df 22; $p=0.19$	20.2 df 21; $p=0.52$
High School Dropout	159 df 24; $p=0$	38.0 df 23; $p=0.03$	31.7 df 22; $p=0.09$	23.9 df 21; $p=0.30$
High School Graduate	193 df 24; $p=0$	57.3 df 23; $p=0.00$	23.8 df 22; $p=0.36$	28.1 df 21; $p=0.14$
College	73.7 df 24; $p=0$	28.3 df 23; $p=0.22$	22.7 df 22; $p=0.43$	27.3 df 21; $p=0.17$

^aIn this table we present tests for zero autocovariance of order 1–4. We provide the test statistic for the hypothesis that the respective autocovariance is zero in all time periods, the degrees of freedom of the test (df), which is determined by the number of time periods for which we can estimate the autocovariance, and the asymptotic p -value for the hypothesis that the autocovariances are zero. We have also tested the hypothesis that all autocovariances of order 3 or higher are jointly zero as in Abowd and Card (1989). The test statistic is 279.83 with 253 degrees of freedom and a p -value of 12%.

Read:Table I reports autocovariances of unexplained earnings growth up to order 4. In the pooled sample they are clearly negative at lags 1 and 2, but close to zero thereafter. Table II formalizes this with tests of zero autocovariances by lag order.

Economic message. The combination of nonzero autocovariances up to lag 2 and near-zero autocovariances at lag 3 or higher implies that the transitory shock is not white noise but is well approximated by an MA(1) process. This short-run persistence is a key input into all later identification steps.

5.3 Table III and Figure 3: permanent shocks are economically important and time varying

TABLE III
THE UNCONDITIONAL VARIANCE OF INCOME SHOCKS^a

	Pooled Sample	High School Dropout	High School Graduate	College Graduate
<i>Conditioning on the Feasible Range of θ</i>				
Transitory shock	0.0439	0.0767	0.0407	0.0160
(upper bound)	(0.0070)	(0.0221)	(0.0079)	(0.0070)
Transitory shock	0.0079	0.0130	0.0077	0.0027
(lower bound)	(0.0012)	(0.0038)	(0.0015)	(0.0012)
Measurement error	0.0303	0.0530	0.0278	0.0114
(upper bound)	(0.0021)	(0.0055)	(0.0030)	(0.0018)
θ (upper bound)	-0.18	-0.17	-0.19	-0.17
<i>Using an External Estimate of σ_ϵ^2</i>				
Measurement error	0.0138	0.0206	0.0129	0.0081
Transitory shock	0.0300	0.0548	0.0267	0.0049
	(0.0031)	(0.0100)	(0.0039)	(0.0030)
θ	-0.2566	-0.2535	-0.2674	-0.5101
	(0.0339)	(0.0610)	(0.0400)	(0.2901)
Permanent shock ^b	0.0313	0.0331	0.0277	0.0437
	(0.0026)	(0.0067)	(0.0039)	(0.0068)

^aEqually weighted minimum distance estimates. Asymptotic standard errors are reported in parentheses.

^bBootstrap standard errors in parentheses.

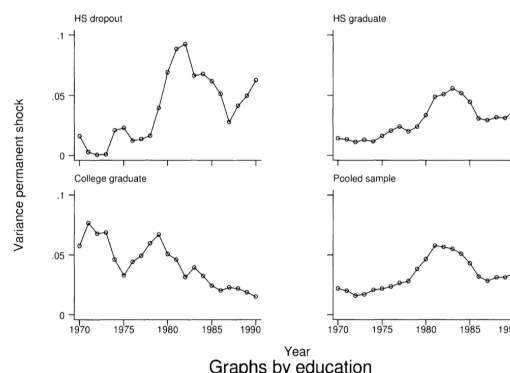


FIGURE 3.—The variance of the permanent shock to earnings.

Read:Table III estimates the unconditional variance of the permanent shock and reports bounds or point estimates for the transitory shock variance, measurement error, and the MA coefficient.

The pooled permanent-shock variance is about 0.0313; the education-specific estimates are about 0.0331 for dropouts, 0.0277 for High School graduates, and 0.0437 for College graduates. Figure 3 plots the time path of the permanent-shock variance.

Economic message. The hypothesis of no permanent shocks is strongly rejected. This matters because permanent shocks are the hard-to-insure component of income risk. Figure 3 then shows that permanent risk itself is not stable over time: it rises through much of the 1970s and early 1980s and then eases after 1984. The College group has the highest unconditional permanent risk in the point estimates.

5.4 Table IV: the within-groups estimator has the identification power

Read:Table IV reports bootstrap χ^2 tests for the relevance of excluded instruments in the reduced forms. The within-groups specification shows very strong relevance; the first-difference specification often does not.

Economic message. This is the paper's most important econometric diagnostic. It justifies the shift from first differences to within-groups GMM and makes the later ARCH estimates much more credible.

	Pooled Sample	High School Dropout	High School Graduate	College Graduate
<i>Transitory Shock</i>				
First Differences				
χ^2 statistic, 2 degrees of freedom	18.84	17.40	26.56	0.96
Bootstrap p -value	0.0000	0.1700	0.3633	0.6233
Within Groups				
χ^2 statistic, 2 degrees of freedom	245.52	42.83	267.88	66.75
Bootstrap p -value	0.0000	0.0033	0.0000	0.0000
<i>Permanent Shock</i>				
First Differences				
χ^2 statistic, 2 degrees of freedom	2.57	1.10	0.10	4.39
Bootstrap p -value	0.2367	0.8833	0.9333	0.2033
Within Groups				
χ^2 statistic, 2 degrees of freedom	360.73	65.31	289.26	86.45
Bootstrap p -value	0.0000	0.0000	0.0000	0.0000

	High School Dropout	High School Graduate	College Graduate
<i>Transitory Shock</i>			
γ^c	0.1918 (0.14)[13%]	0.6672 (0.152)[0.4%]	0.3878 (0.192)[10.3%]
Age	-0.0030 (0.0065)	-0.0005 (0.0029)	-0.0006 (0.0027)
Age ² /100	0.0028 (0.0082)	0.0010 (0.0036)	0.0007 (0.0036)
P -value OID test ^b	19%	14%	15%
P -value for time effects	95%	99%	99%
P -value for time and age effects	98%	96%	98%
P -value test for unobs. heterogeneity ^c	34.4%	2%	4.3%
γ^c Pooled (all education groups) ^d		0.404 (0.096)[0%]	
<i>Permanent Shock</i>			
φ^c	0.3299 (0.21)[7.4%]	0.8912 (0.18)[0.0%]	0.0283 (0.33)[90.4%]
Age	-0.0076 (0.0056)	0.0017 (0.0034)	-0.0065 (0.0055)
Age ² /100	0.0090 (0.0063)	-0.0008 (0.0040)	0.0083 (0.0066)
P -value OID test ^b	14%	40%	2%
P -value for time effects	93%	89%	95%
P -value for time and age effects	93%	93%	91%
P -value test for unobs. heterogeneity ^c	0%	2.7%	37.2%
φ^c Pooled (all education groups) ^d		0.56 (0.126)[0.3%]	

^aBootstrap standard errors based on 1000 replications reported in round brackets and p -values in square brackets.

^bThe OID test is a Sargan test for the null of instruments validity. It is distributed χ^2 with degrees of freedom equal to the number of exclusion restrictions.

^cThe test for the null of no unobserved heterogeneity is based on 500 replications.

^dThe pooled estimate and its standard error are obtained by minimum distance.

5.5 Table V: state dependence in risk is strong, especially for permanent shocks

Read:Table V reports the ARCH coefficients for the conditional variances. For the pooled sample, the transitory-shock coefficient is about 0.404 and the permanent-shock coefficient is about 0.56. By education group, the largest point estimates are for High School graduates: about 0.6672 for

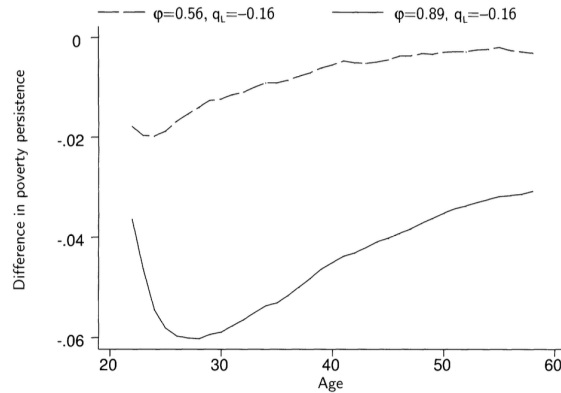


FIGURE 4.—Difference in the persistence of poverty between the ARCH and the i.i.d. case.

transitory shocks and about 0.8912 for permanent shocks.

Age and time effects are mostly insignificant once the lagged variance proxy is included. The table also reports evidence of unobserved heterogeneity in variances for several education-shock combinations.

Economic message. The central message is that volatility clusters. This is especially strong for permanent shocks, which are the most consequential for lifetime resources and insurance. The heterogeneity tests also say that some workers are persistently more exposed to risk than others, even after conditioning on observables.

5.6 Figure 4: ARCH changes the implied persistence of poverty

Read:Figure 4 compares poverty persistence under the estimated ARCH income process with poverty persistence under a counterfactual i.i.d.process. The plotted difference is negative over the life cycle.

Economic message. ARCH does not simply “increase persistence everywhere.” In the tails of the income distribution it can reduce persistence, because people with unusually large shocks are also the people with unusually high conditional variance next period. They are therefore more likely to move out of the tail. The effect is strongest when the permanent-shock ARCH coefficient is larger.

6 Short summary

- Estimate a flexible earnings process for PSID men with education-specific mean dynamics and three latent residual components: permanent shocks, transitory shocks, and measurement error.
- Use autocovariances of earnings growth to show that the transitory component is MA(1) and that permanent shocks are definitely present.
- Model the conditional variance of both permanent and transitory shocks as ARCH(1) processes with time effects and permanent individual heterogeneity.
- Use long-panel moment conditions and within-groups GMM to identify the ARCH coefficients.

- Find strong state dependence in risk: pooled ARCH persistence is roughly 0.404 for transitory shocks and 0.56 for permanent shocks.
- Find additional evidence that earnings variance differs permanently across individuals.
- Show that these variance dynamics matter economically for poverty persistence and precautionary savings.

7 Conclusion

The paper's central achievement is to show that the stochastic process for earnings is richer than the standard homoskedastic framework used in many quantitative models. Earnings are driven by both permanent and transitory shocks, and the variances of both shocks are themselves persistent and heterogeneous.

Methodologically, the paper contributes an identification strategy that works even when the permanent and transitory shocks are not separately observed, the transitory component is serially correlated, and measurement error is present. Empirically, the long PSID panel is rich enough to support these moments and to reject the simpler i.i.d. benchmark.

The most important economic lesson is that microeconomic uncertainty is dynamic. A worker who was hit by a large earnings shock yesterday is not just poorer today; he is also more exposed to volatility tomorrow. That fact matters for consumption smoothing, savings behavior, poverty dynamics, and any model that tries to map labor-income risk into household decisions.